Workbook · Hand calculation

State Space Hand Calculations

Follow one Kalman filter step without a calculator

Spending · Rates · EWC–EWA hedging · Stochastic volatility · EM · Smoother

Using the hypothetical numbers from the main deck, we work through predict → compare → update line by line.

Every hand calculation uses the same seven lines

Case where the state and the observation are single numbers

$$\hat{x}_t^- = A\hat{x}_{t-1} \qquad \text{state forecast}$$

$$P_t^- = A^2 P_{t-1} + Q \qquad \text{forecast error variance}$$

$$e_t = y_t - H_t\hat{x}_t^- \qquad \text{innovation (actual - expected)}$$

$$S_t = H_t^2 P_t^- + R \qquad \text{innovation variance}$$

$$K_t = \frac{P_t^- H_t}{S_t} \qquad \text{Kalman gain}$$

$$\hat{x}_t = \hat{x}_t^- + K_t e_t \qquad \text{state update}$$

$$P_t = (1 - K_t H_t) P_t^- \qquad \text{updated error variance}$$

Order of work

- Predict: move state and error variance one step ahead
- Compare: innovation and its variance
- Update: fix state and error variance with the gain

Spending example: from 300 to 306

A = H = 1 · unit 10,000 KRW · variance unit (10,000 KRW)^2

Item	Value
Last month's estimate	300
Last month's error var.	8
True change variance Q	8
This month's survey	312
Survey error variance R	16

Next month

- Start again from 306 and 8

1. Predict $\hat{x}_t^- = 300, \qquad P_t^- = 8 + 8 = 16$

2. Compare $e_t = 312 - 300 = 12, \qquad K_t = \frac{16}{16 + 16} = 0.5$

3. Update state $\hat{x}_t = 300 + 0.5 \times 12 = 306$

4. Update var. $P_t = (1 - 0.5) \times 16 = 8$

Rate example: from 3.00% to 3.05%

A = H = 1 · unit % · variance unit (%p)^2

Item	Value
Level forecast	3.00
Forecast error var.	0.0025
Obs. residual var. R	0.0075
Observed rate	3.20

1. Kalman gain

$$K_t = \frac{0.0025}{0.0025 + 0.0075} = 0.25$$

2. Update state

$$\hat{\ell}_t = 3.00 + 0.25(3.20 - 3.00) = 3.05\%$$

Check

- The rate came in 0.20%p higher and the gain is 0.25, so the level rises 0.05%p

Hedging example: hedge ratio from 0.80 to 1.00

The link number is the EWA return 0.010, so the gain can exceed 1

Item	Value
Hedge ratio forecast	0.80
Forecast error var.	0.04
EWA return	0.010
Actual EWC return	0.012
Obs. residual var. R	0.000004

1. Expected return $\hat{y}_t = 0.80 \times 0.010 = 0.008$

2. Innovation $e_t = 0.012 - 0.008 = 0.004$

3. Kalman gain $K_t = \frac{0.04 \times 0.010}{0.010^2 \times 0.04 + 0.000004} = 50$

4. Update ratio $\hat{\beta}_t = 0.80 + 50 \times 0.004 = 1.00$

5. Update var. $P_t = (1 - 50 \times 0.010) \times 0.04 = 0.02$

Check the hedge ratio in dollars and P&L

Ratio 0.8 · hypothetical case where both ETFs follow the relation

Position	Return	P&L calculation	P&L
Long EWC $100	+0.8%	100×0.008	+$0.8
Short EWA $80	+1.0%	-80×0.010	−$0.8
Total			$0

Converting to share counts

$$\text{EWA shares to short} = \beta \times \text{EWC shares held} \times \frac{\text{EWC price}}{\text{EWA price}}$$

Stochastic volatility example: from 1% to about 1.25%

Transformed observation equation · observation error variance 4.9348 · forecast error variance 1

Item	Value
Vol forecast	1%
Today's return	+2%
Log constant	−1.2704
Obs. error var.	4.9348
Forecast error var.	1

1. Forecast state

$$\hat{h}_t^- = \log(0.01^2) \approx -9.2103$$

2. Transformed obs.

$$z_t = \log(0.02^2) + 1.2704 \approx -6.5536$$

3. Kalman gain

$$K_t = \frac{1}{1 + 4.9348} \approx 0.1685$$

4. Update state

$$\hat{h}_t = -9.2103 + 0.1685[-6.5536 - (-9.2103)] \approx -8.7627$$

5. Volatility

$$\hat{\sigma}_t = \exp(\hat{h}_t/2) \approx 1.25\%$$

One EM round: average E-step outputs into a new R

※ three illustrative months · real EM uses the full record and builds the new Q the same way

Mo.	Survey	Smoothed	Uncert.
1	312	308	6
2	305	304	5
3	298	300	6

Where the values come from

- The smoothed mean and uncertainty are computed by the E-step with the current Q and R

E output · month 1

$$(312 - 308)^2 + 6 = 22$$

E output · month 2

$$(305 - 304)^2 + 5 = 6$$

E output · month 3

$$(298 - 300)^2 + 6 = 10$$

M output · new R

$$R_{\text{new}} = \frac{22 + 6 + 10}{3} \approx 12.7$$

One smoother step: refine last month 300·8 to 303·6

Spending numbers · this month is the last; step back one period

Item	Value
Last month filtered	300 · 8
This month forecast	300 · 16
This month updated	306 · 8

Intuition

- Seeing this month rise, raise last month too; having seen more data, cut its uncertainty

1. Start $m_T = 306, \qquad V_T = P_T = 8$

2. Ratio $J = \frac{8}{16} = 0.5$

3. Mean $m_{T-1} = 300 + 0.5 \times (306 - 300) = 303$

4. Variance $V_{T-1} = 8 + 0.5^2 \times (8 - 16) = 6$